SIMATS ENGINEERING

SIMATS Online Programming Lab

C PROGRAMMING

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QUESTIONS 25 / 25 completed

Inverse

INVERSE

Write a C program to find the inverse of a 2x2 matrix.

PROGRAM EDITOR

C | Run | Save

```c
#include <stdio.h>

int main() {
    float mat[2][2], inv[2][2];
    float det;
    int i, j;

    // Read elements of the 2x2 matrix
    printf("Enter the elements of the 2x2 matrix:\n");
    for (i = 0; i < 2; i++) {
        for (j = 0; j < 2; j++) {
            scanf("%f", &mat[i][j]);
        }
    }

    // Calculate the determinant
    det = (mat[0][0] * mat[1][1]) - (mat[0][1] * mat[1][0]);

    if (det == 0) {
        printf("Inverse is not possible because the determinant is
    } else {
```

OUTPUT

```
Enter the elements of the 2x2 matrix:
Inverse of the 2x2 matrix is:
0.60 -0.70
-0.20 0.40
```

INPUT

```
4 7
2 6
```

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